Canonical Quantization of the SAT Field Sector

I. CANONICAL QUANTIZATION OF THE SAT FIELD SECTOR

The derivation of the quantum properties of the Scalar–Angular Theory (SAT) proceeds through the canonical quantization approach applied to the continuous field degrees of freedom [1]. The formulation relies on the canonical commutation relations established for the field and its conjugate momentum over the continuous sector [2].

A. The General Canonical Formalism

The dynamics of the SAT continuous fields $(\theta_\mu, u_\mu, J_{\mu\nu})$ [3], governed by the total Lagrangian density $\mathcal{L}_{\text{total}}$ [4], require the definition of canonical conjugate momenta π corresponding to the generalized fields ξ [5].

For the gravitational component of the SAT framework, the Hamiltonian density $\mathcal{H}_{\text{GR+SAT}}$ is obtained via a Legendre transform, resulting in the Hamiltonian and momentum constraints $(\mathcal{H}_0, \mathcal{H}_i^*)$ expressed using the ADM lapse N and shift N^i variables [6]. This approach involves the canonical momentum π^{ij} conjugate to the spatial metric h_{ij}^* [7].

B. Canonical Quantization Condition

In the general continuous sector of SAT, canonical quantization proceeds through the definition of symbolic commutators [1]. If $\xi(\lambda)$ represents a continuous field component and $\pi(\lambda')$ represents its canonical conjugate momentum [5], the quantization condition is given by the commutation relation:

$$\boxed{[\xi(\lambda), \pi(\lambda')] = i\hbar\,\delta(\lambda - \lambda')}. \tag{1}$$

This relation establishes the standard quantization condition over the continuous sector of the theory [2]. The relationship holds over the continuous variable λ, signifying that the commutator is non-zero only when the field locations coincide $(\lambda = \lambda')$ [2, 5].

[1] Placeholder reference 1.
[2] Placeholder reference 2.
[3] Placeholder reference 3.
[4] Placeholder reference 4.
[5] Placeholder reference 5.
[6] Placeholder reference 6.
[7] Placeholder reference 7.